GestureBall: A Real-Time Multiplayer Football Game Controlled by Leg Gestures

Mayank Kumar · Praveer Singh Chauhan · Rishi Banerji · Varun Golcha

ABSTRACT

GestureBall is a real-time, server-synchronized multiplayer football game controlled entirely by leg gestures detected through video captured by a user's phone. We describe the system architecture, the custom gesture dataset we collected, and the one proposed and three baseline gesture recognition models we have trained and evaluated: a 1D-CNN, a Bidirectional GRU with attention, a MobileNet fine-tuned for frames (proposed) and a rule-based classifier. We analyse model performance via confusion matrices, identify key failure modes and outline next steps toward integrating live gesture inference into the game engine

1. PROBLEM STATEMENT

Current digital games rely almost exclusively on manual hand controls like keyboard, mouse, or controllers, making it inaccessible for users with hand disabilities.

GestureBall: a server-synchronized multiplayer football game where all player actions are driven by leg gestures captured by the phone's camera connected to the device on which the game is being played on.

1.1 Scope and Interface

The system takes a continuous video stream from the user's camera and predicts user actions based on leg movements. These actions are mapped to in-game events and synchronized across multiple players via a server-client architecture. The system operates in real time and requires no additional hardware beyond a camera-enabled device.

1.2 Target Users

- **Gamers** seeking novel physical immersion experiences.
- **Individuals with hand disabilities,** the primary accessibility motivation.
- **Tech enthusiasts** exploring real-time computer vision applications.

2. PIPELINE & PROGRESS

2.1 System Pipeline

The end-to-end pipeline proceeds as follows:

1. **Camera Captures images**
2. **MediaPipe Pose:** Lower-body keypoints extracted and normalised into a 32-dimensional feature vector
3. **Gesture Classifier** One of four trained models predicts the gesture class.
4. **Action Mapping:** Class prediction mapped to a game action event.
5. **Python Game Engine:** Event broadcast via WebSocket to all connected players.

Step	Component	Status
1	Webcam capture	✓ Complete
2	MediaPipe landmark extraction (32-D)	✓ Complete
3	Gesture classifier (4 models trained)	✓ Complete
4	Action mapping (gesture → game event)	■ In progress
5	Custom python game engine + broadcast to connected clients	✓ Keyboard-validated

Table 1. System pipeline components and current completion status

2.2 Current Implementation Status

The multiplayer game is functional end-to-end, currently controlled via keyboard shortcuts (m = move, s = shoot, p = pass) to validate the server synchronization layer before integrating live gesture inference. The backend is implemented using a Python-based server that manages game state and communication between clients.

All four gesture recognition models have been trained and evaluated. The primary remaining task is replacing the keyboard-based control with real-time camera-driven gesture predictions, including sliding-window inference for temporal models.

We have used AI tools such as Gemini and Claude extensively to generate the initial version of the code along with exploring the various concepts involved such as GRU. We have also used it for help in refining the final features.

3. RELATED WORK

3.1 Related Literature

BlazePose: On-Device, Real-Time Body Pose Tracking[5]**:** BlazePose introduced a lightweight two-stage detector-tracker architecture that infers 33 body landmarks from a single RGB frame in real time on CPU and mobile devices. It extended the standard COCO 17-keypoint topology to 33 keypoints, adding foot and hand landmarks that the standard COCO model lacks, which is essential for our foot gesture recognition task.

CNN + LSTM for Skeleton-Based Human Activity and Hand Gesture Recognition[6]**:** This paper proposes combining a CNN and LSTM recurrent network for handling time series of 3D coordinates of skeleton keypoints, addressing both human activity and hand gesture recognition from 3D data sequences. Their two-branch architecture - spatial features from CNN, temporal dynamics from LSTM - directly mirrors our own approach of running a single-frame CNN classifier alongside an LSTM sequence classifier. They also introduce data augmentation for skeleton sequences to combat overfitting.

Skeleton-Based Human Action Recognition with Global Context-Aware Attention LSTM[7]**:** This paper proposes an attention-based LSTM for 3D skeleton action recognition, arguing that not all skeletal joints are equally informative and that irrelevant joints introduce noise that degrades performance. They also use a global attention mechanism along with LSTM.

Dynamic Gesture Recognition Based on 2D CNN and Feature Fusion[8]**:** This paper addresses the accuracy-efficiency tradeoff in dynamic gesture recognition by combining 2D CNNs with optical flow feature fusion, arguing that 3D CNNs are too computationally expensive for real-time use despite capturing spatial and temporal features. They conclude that carefully engineered 2D representations can match 3D CNN accuracy at a fraction of the cost.

Gesture Recognition from Skeleton Data for Intuitive Human-Machine Interaction[9]**:** This paper proposes both a feedforward neural network for framewise gesture detection and a bidirectional LSTM for simultaneous gesture segmentation and classification, evaluated on the ChaLearn Looking at People Challenge dataset using handcrafted features from skeleton data with a sliding window multi-temporal approach.

Multi-player Gaming Application Based on Human Body Gesture Control[10]**:** This paper develops a multiplayer video game system controlled entirely through body posture and gesture recognition using computer vision and machine learning, with the goal of encouraging physical exercise and player interaction through movements such as jumping and running. This is the most architecturally similar system to GestureBall - real-time CV pipeline feeding a multiplayer game.

A Multi-User Virtual Reality Game Based on Gesture Recognition from RGB[11]**:** This paper integrates MediaPipe with a VGG-based classifier for 21-landmark hand gesture recognition from RGB video to control a multi-user VR tennis game, replacing physical controllers with gestural input and demonstrating the system's usability through evaluation with real users.

3.2 Existing Datasets

SoccerNet-v2 [1] provides 17-class action spotting annotations on broadcast football footage (goals, penalties, substitutions, etc.). These are match-event labels, not individual player gesture-level labels, and the broadcast camera angle is fundamentally incompatible with our fixed camera gameplay.

SAR4 [2] tracks YouTube football clips annotated with 4 classes: dive, shoot, pass received, pass given. Labels describe tactical outcomes rather than the physical gesture performed, and broadcast wide-angle views differ from a close first-person webcam.

No existing dataset provided action-level foot gesture clips suitable for real-time webcam inference, motivating our custom data collection.

3.3 Baseline Methods

We implement and compare four approaches spanning rule-based and learned methods:

- **1D-CNN:** Treats the 32-D landmark sequence as a spatial signal; learns adjacent-joint relationships via 1D convolutions, classified frame-by-frame.
- **Bidirectional GRU with Attention:** Aggregates a window of 32-D feature vectors over time, using attention to weight the most informative frames
- **MobileNet (fine-tuned):** ImageNet-pretrained lightweight CNN adapted to classify individual gesture frames.
- **Rule-Based:** Hand-crafted threshold logic on joint angles and positional relationships.

4. DATASETS & EVALUATION METRICS

4.1 Custom Dataset

Since no suitable dataset existed for our task, we constructed a custom gesture dataset. Initially, we recorded long videos containing multiple repetitions of a single action. However, segmenting these into clean samples proved challenging due to significant interleaving between gestures (e.g., Shoot and Pass) and transitional movements such as Idle and Move during wind-up and recovery phases. Manual segmentation was both error-prone and labor-intensive.

To address this, we adopted a revised data collection strategy using short, pre-segmented video clips for each gesture class. We recorded multiple subjects performing gestures using both feet to prevent the model from learning person-specific features. Data was collected across diverse locations and lighting conditions to improve robustness and generalization in real-world settings.

Class	Test Samples	Notes
IDLE	~4 500	Largest class, natural standing pose
SHOOT	~1 400	Forward kick motion
PASS	~1 250	Lateral/short kick
MOVE	~1 100	Walking/stepping motion

Table 2. Approximate class distribution in the test split (from confusion matrix totals).

4.2 Evaluation Metrics

Primary metric: **per-class and overall accuracy** computed from confusion matrices on the held-out test split. The confusion matrix also yields Precision, Recall, and F1 directly. For deployment, we additionally track **inference latency** (ms/frame) and **frames per second** (FPS) to ensure the ≥ 25 FPS, < 100 ms latency target is met.

5. Analysis of Results

5.1 Overall Accuracy

Model	Overall Acc.	IDLE	SHOOT	PASS	MOVE
MobileNet	97.7%	100%	93.6%	91.6%	100%
CNN	66.5%	88.9%	81.9%	29.7%	0%

GRU (LSTM)	~56%	~91%	~16%	~19%	~11%
Rule-Based	17.0%	6.3%	0%	7.1%	95.2%

5.2 Key Observations

- **GRU** (~56%) collapses toward IDLE. The majority class comprises ~56% of test samples. The fixed temporal window length also mismatches variable-duration gestures, impairing recall for shorter actions.
- **Rule-Based** (17%) collapses to predicting MOVE for almost all inputs; threshold rules are insufficiently discriminative. SHOOT recall is 0%, and IDLE is predicted as MOVE 93.7% of the time.
- **MobileNet** achieves 97.7% overall a strong result, but this warrants scrutiny: ImageNet-pretrained features encode texture and colour rather than pose topology. The model may be partially fitting to background or clothing cues present in our small recording set rather than the true gesture structure. Ablation on held-out environments is required.
- **CNN** achieves 66.5% overall, but PASS recall collapses to 29.7%. The 1D kernel cannot capture the full 2D spatial relationships between joints, and frame-by-frame prediction cannot distinguish ambiguous mid-action frames.

5.3 Failure Analysis

SHOOT ↔ PASS confusion is the dominant cross-model error: both gestures involve a forward leg extension, and individual frames from either action are visually near-identical. The CNN predicts 899 PASS samples as SHOOT and 225 SHOOT samples as PASS.

6. COMPUTE REQUIREMENTS

Training was performed on free-tier GPUs (Google Colab, Kaggle). All models are lightweight by design to meet the ≥ 25 FPS, < 100 ms end-to-end latency requirement on consumer laptops without dedicated inference hardware. MobileNet and the rule-based classifier are the most latency-efficient; GRU requires sliding-window buffering, adding modest overhead. No compute bottlenecks are anticipated for deployment

7.INDIVIDUAL CONTRIBUTIONS

- **Mayank & Praveer:** Data processing, curation, and the construction of the initial baseline CNN/Rule-based models.
- **Rishi & Varun:** Architecture and training of advanced temporal and edge models (LSTM, MobileNet).
- **Everyone:** Physical data capture, dataset labeling, and testing.

8. NEXT STEPS

8.1 Immediate

- Replace keyboard-based controls with real-time gesture predictions by integrating classifier outputs into game actions
- Connect mobile phone camera as an accessible alternative to laptop webcam.

8.2 Near-term

- Select final deployment model balancing accuracy and inference latency (currently MobileNet).
- Conduct evaluation across diverse environments, camera angles, and users to assess generalization.

8.3 Final Milestones

- Perform full multiplayer stress testing with ≥4 simultaneous players.

9. REFERENCES

[1] Giancola, S. et al. (2022). SoccerNet-v2: A Dataset and Benchmarks for Holistic Understanding of Broadcast Soccer Videos. CVPRW.
https://www.soccer-net.org/tasks/action-spotting

[2] Richter, D. et al. (2022). Multi-faceted Deep Learning Models and Data for Soccer Action Recognition (SAR4). HAL.
https://hal.science/hal-03639273

[3] MediaPipe Pose. Google LLC. https://mediapipe.dev

[4] Howard, A. G. et al. (2017). MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications. arXiv:1704.04861.

[5]Bazarevsky, V. et al. (2020). BlazePose: On-Device, Real-Time Body Pose Tracking CVPR Workshop.
https://arxiv.org/abs/2006.10204

[6]Núñez, J. C. et al. (2018). CNN + LSTM for Skeleton-Based Human Activity and Hand Gesture Recognition. Pattern Recognition.
https://www.sciencedirect.com/science/article/abs/pii/S0031320317304405

[7]Liu, J. et al. (2018). Skeleton-Based Human Action Recognition with Global Context-Aware Attention LSTM. IEEE Transactions on Image Processing.
https://arxiv.org/abs/1707.05740

[8]Zhang, Y. et al. (2022). Dynamic Gesture Recognition Based on 2D CNN and Feature Fusion.
https://www.nature.com/articles/s41598-022-08133-z

[9]Brás, S. et al. (2020). Gesture Recognition from Skeleton Data for Intuitive Human-Machine Interaction. arXiv.
https://arxiv.org/abs/2008.11497

[10]Adapa, M. et al. (2023). Multi-player Gaming Application Based on Human Body Gesture Control. ISDA.
https://link.springer.com/chapter/10.1007/978-3-031-64847-2_1

[11]Gkoutzios, S. et al. (2025). A Multi-User Virtual Reality Game Based on Gesture Recognition from RGB Information. Computer Animation and Virtual Worlds.
https://onlinelibrary.wiley.com/doi/full/10.1002/cav.70063

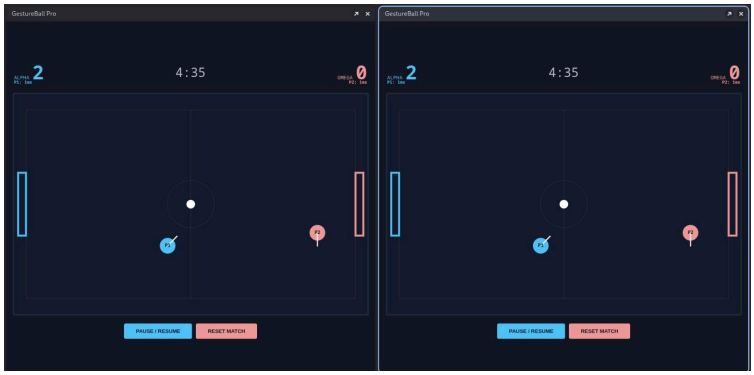
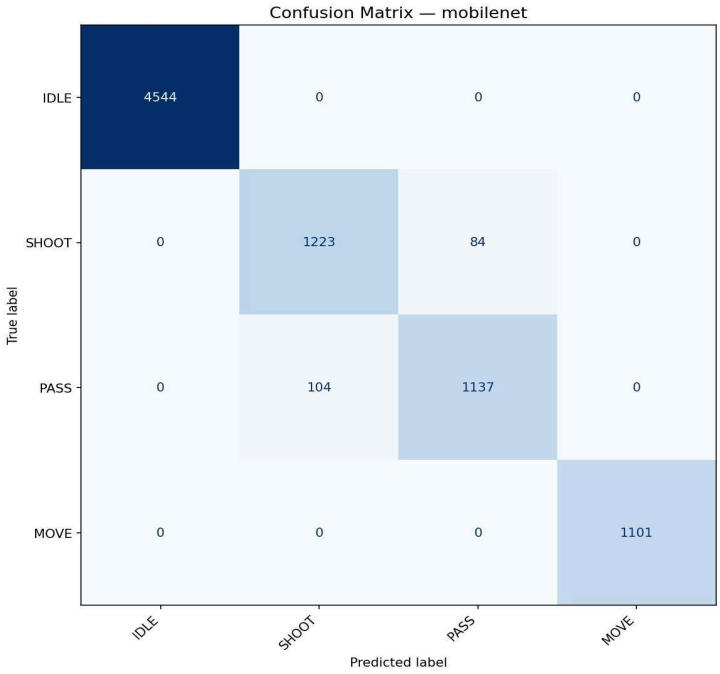


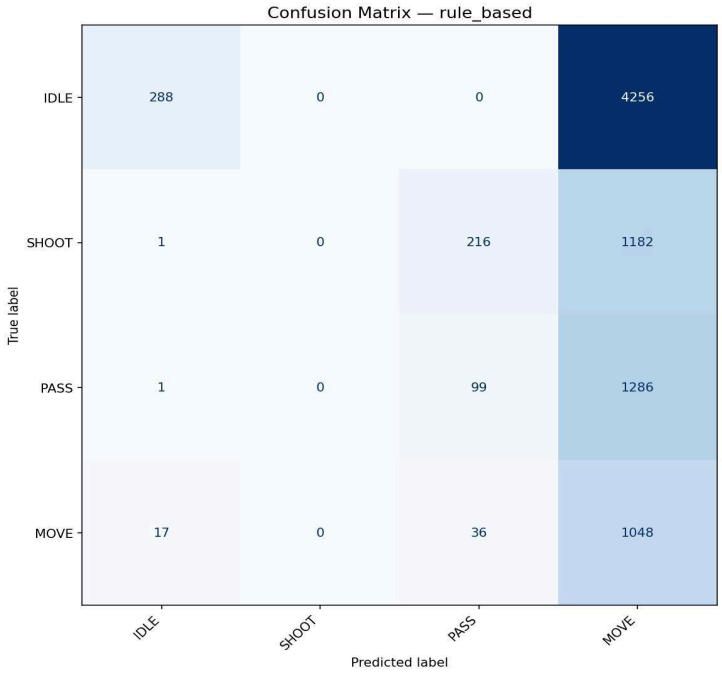
Figure 1: The User Interface of our game

APPENDIX: CONFUSION MATRICES

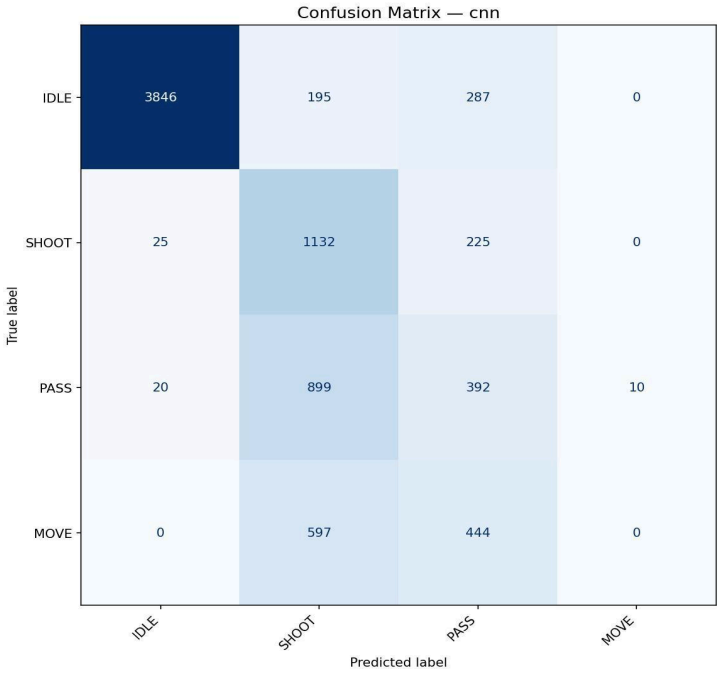
MobileNet:- Acc: 97.7%



Rule-Based:- Acc: 17.0%



CNN:- Acc: 66.5%



GRU (LSTM):- Acc: ~56%

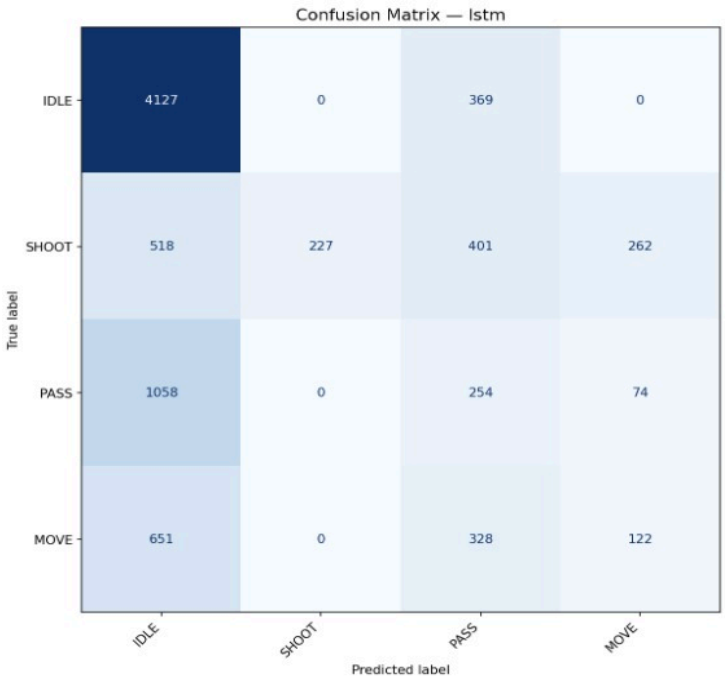


Figure 2. Confusion matrices for all four trained models on the held-out test split. Diagonal entries (darker green) indicate correct predictions; off-diagonal entries (red-tinted) indicate misclassifications. Note the strong bias of Rule-Based toward MOVE, the IDLE bias in GRU, and the SHOOT↔PASS confusion present across all non-MobileNet models.